Ferdinan Nathanael

Undergraduate at Binus University

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SUMMARY

Highly motivated Computer Science student with a strong interest in software development, database systems, software analysis, and web development. Skilled in database integration, structured data management, backend system development, and performance monitoring. Experienced in working with relational databases, SQL queries, with a strong eagerness to gain hands-on industry experience and collaborate with experienced mentors. Currently an intern at PLN ICON PLUS with focus on chatbot building and database maintenance.

PROJECTS

Sports matchmaking app — Jan 2024-Mar 2024

- Implemented matchmaking logic inspired by the Americano format, ensuring fair player distribution and competitive match results.
- Developed input forms for entering player data and skill ratings, enabling quick generation of team matchups without account registration.
- Utilized PostgreSQL to store player lists and match results, enabling efficient recalculation and data retrieval.
- Designed a responsive front-end interface to display match pairings and game sequences clearly for participants and organizers.

WhatsApp Sales Automation Bot (PLN ICON PLUS) — Feb 2026-Ongoing

- Developed a WhatsApp-based sales automation bot using FastAPI and webhook integration to handle customer inquiries and automate initial sales conversations.
- Implemented multi-step conversation flow and state management with SQLite, enabling the bot to collect customer needs, company details, and service requirements in a structured way.
- Built a recommendation logic system to suggest suitable company prospects or services based on customer input and internal business data.
- Integrated a knowledge bank using ChromaDB, PDF processing, embeddings, and AI-generated responses to answer product-related questions with reduced hallucination.
- Designed backend endpoints for receiving, processing, and responding to WhatsApp messages through Botika webhook integration.
- Improved sales workflow efficiency by reducing repetitive manual responses and preparing structured customer information for follow-up by the sales team.

Billiard Matchmaking App — May 2026-ONGOING

- Built a full-stack pool-hall platform covering player check-ins via QR, match invitations, score confirmation, payment-gated settlement, tournaments, and role-based admin tooling
- Designed a security-first backend with 200+ SQL migrations, row-level security policies, idempotent RPCs, and 120+ pgTAP database tests

SKILLS

Front-end: HTML, CSS, JavaScript, React.js, Bootstrap, Tailwind CSS
Back-end: Node.js, Java, Python, PHP, Laravel, C#, ASP.NET
Database: PostgreSQL, MySQL, MongoDB, SQL Server
UI/UX: Figma, Canva,
Business analyst: proficient in agile programming method
System analysis: UML, Visual Paradigm

EDUCATION

Computer Science Undergraduate at Binus University (2023-ongoing)
Intern at PLN ICON PLUS (2026-ongoing)

Relevant Coursework: Database Engineering, Data Structure, Web Programming, Agile Software Development, code Reengineering, Software Engineering, Artificial Intelligence, Framework Layer Architecture, Pattern Software Design, Object Oriented Analysis and Design, Operating System